The model family and the landscape

Sleeping Machines is a family of causal networks of stateful temporal programs. Content, computational time, addresses and persistent evidence jointly determine computation. A common interface supports different programs and mixtures at every level.

The envelope supplies choices; a member fixes compatible operators and contracts; the integrated target is measured on temporal/selective quality and resource behavior.

A generative design space, from atoms to systems

Choose evidence, flow, writes, keys, reception, output and credit; then compose modules, interactions, layers, stacks and complete learning systems. Uniform width, fan-in and scheduling are optional design choices.

Review computational power and total work, represented distinctions, and useful learning at each level. Expand the earliest limiting region: downstream richness cannot recover erased evidence.

The same three axes at every level

Computational power and required work, represented distinctions, and useful learning are separate properties, and each must survive the interfaces that compose a model.

Level	Power / work	Representation	Trainability
Atoms	Flow, delays, gates, races, sums and their ordering	Content, order, interval, phase and repeated evidence	Continuous sensitivity plus event-boundary utility
Units	State transformation, write, emission and readout	Retained facts, temporal modes and message bandwidth	Content/time/state credit; conditioning and exposure
Modules	Discovery, candidate proposals, reception and commits	Accessible evidence and which messages can meet	Supported alternatives, selection and write utility
Layers	Mixing, shared reads, transport and timed joins	Relations across programs, sources and values	Coupled derivatives and changed event histories
Stacks	Depth, recurrence and optional continuation	Hierarchical features and persistent context	Information paths, Jacobian products and writer horizon
Compositions	Neural, statistical, historical and rich local regions	Abstraction plus direct evidence at a common query	Fusion responsibilities, retrieval and future utility
Systems	Schedule, caches, batching, optimizer and serving	Usable state within information/resource limits	Causal objectives, versions, update consistency and cost

Diagnose lost information, unsupported computation and missing credit separately. Then test their coupled consequence in an integrated member, preserving the successful parent and the full resource boundary.

Relationships to established model families

The envelope intersects existing categories and contains reference constructions for them. Its contribution is an economical integration of time, state, selection and learning; existing systems can also be extended or hybridized.

Family	What can be included	Additional design freedom	Condition / price
Recurrence / SSM	Supported state transitions and filters	Addressed private programs and timed nonlinear interaction	Actual operators, retained state and stable learning
Attention / Transformer	Supported attention and full reference blocks	Persistent evidence, races and variable delivery fan-in	Full aggregation, FFN/residual/norm/positions; all required work
Sparse experts / MoE	Supported gating and expert mixtures	Private stateful experts, clocks and future write utility	Candidate discovery, mixture aggregation and trained specialization
Counts / retrieval	Causal estimators and explicit historical evidence	Shared learned abstractions and temporal/nonlocal fusion	Correct statistics, support, access, updates and storage
Adaptive memory / TTT	Specified inner learner and causal deployment updates	Selective temporal programs and local learning policies	Actual objective/update, versions, retained credit and extra work
Event / temporal	Local event scheduling, timers and supported temporal logic	Small vector messages and coupled alternative-route credit	Precision, ordering, termination, traffic and measured hardware

A function inclusion is not a trained-quality or resource result. A restricted winner-only member has less interaction support than a full-aggregation member. Broader operator choices provide a route to matching needed functions without promising that present native units already implement every reference.

Units combine content, timing and persistent evidence

The native receiver below illustrates one branch. It combines a stored key, input-conditioned temporal proposal, small residual value and computational clock. Keys choose access; values carry content; the selected write changes future evidence.

A unit is a product of choices: statistical fields, neural vectors, protected facts or fast parameters; identity, decay, rotation or richer local maps; first, multi-message, window or silence reception. Their integration changes the function and the credit needed to learn it.

Modules, interactions, layers and stacks

Candidate discovery, winner selection, reception and continuation are distinct decisions. Heads expose complementary evidence only through declared mixing, joins or shared reads. The diagram is the native branch, not the whole family definition.

Depth creates hierarchical temporal functions when information and credit survive interfaces. Persistent facts, accessible facts and trainable writers are separate capacities. Shared maps do not automatically share private state across sources.

One family across dense and sparse worlds

Dense frames, language, queries and irregular events can feed explicit shared information paths. Input density, computation density, arrival cadence and physical clocking are independent choices.

A rich local island can reproduce supported attention or other reference operators while surrounding regions remain selective. Full participation needs actual aggregation; synchrony needs snapshots/barriers. Joint multimodal training remains an integration ambition. No global periodic tick is required; physical clockless energy is unmeasured.

The same structure can learn, batch and adapt

Representation, timing, routes and writes form one coupled credit problem. Factual derivatives teach a realized history; useful alternative returns teach changed routes, reception and later state consequences. The successful local value teacher does not cover full future-write utility.

An earlier neural family pilot improved causal online prediction at 10.736 times total processing work. It is scoped evidence, not current native-backbone TTT or drift retention. State/statistical adaptation is distinct from gradient TTT. Local, streamed and batched updates need ownership and versions. General morphing and fully asynchronous hardware training remain open; learned event routes already exist.

A precise core with contrasting complete members

A member declares the following twelve semantic fields. A construction grammar permits serial, parallel, recurrent, memory-linked, scheduled and policy-controlled regions, subject to compatibility and termination.

Field	Required contract
E / Evidence	Observation schema, provenance, time/address adapters and admissible histories
O / Operators	Actual maps, domains, precision and rounding for numerical claims
G / Composition	Units, connections, shared paths and structural-action space
S / State	Facts, timing, queues, caches, randomness, ownership and resets
I / Interfaces	Content/address/time types, conversions and state access
P / Participation	Discovery, keys, proposals, deliveries, writes and alternative support
T / Schedule	Order, delays, ties, deadlines, joins, cancellation and barriers
Q / Queries	Observed cutoff, accessible snapshot, completion and readout
L / Objectives	Task/self-supervision, labels, scoring order and resource penalties
U / Learner	Credit, horizon, optimizer/update ownership and versions
B / Bounds	Finite execution, capacities, admission and overflow policy
X / Execution	Reference/deployment equivalence, measurement and evidence status

C: causal binary counts, addressed increments and a full prediction distribution; no gradient TTT. R: two private temporal programs, hard race and a local online value teacher; no deep or full writer credit. H: shared sparse reception, two-record dense query and exact finite-option episode TTT. These complete illustrative specifications share one template; they are not new fitted models or benchmark claims.

Composition and adaptive growth have separate contracts

Local capability reaches the system only through compatible information, state, time, query, credit and resource interfaces. Serial, parallel, recurrent and hybrid compositions expose different evidence and dependencies.

Forward preservation checks future state and emissions, including pending deadlines. Learning preservation additionally checks credit and parameter/optimizer migration. Resource refinement checks actual work, storage, traffic and latency. Equal current predictions do not establish any of these stronger claims. Intended learned deviations remain permitted.

Adaptive computation keeps the task and query contract fixed

Learned routes, delays, reception and stopping change internal execution. Ordinary quality comparisons still score the same declared targets. If a branch is unfinished, specify the accessible snapshot and fallback or mark the run invalid. Internal silence need not mean the system declines to answer.

Choice	Query / learning obligation	Full resource boundary
Slow route / deadline	Name the clock; use the declared snapshot/fallback; retain the target	Attempted work, pending state and permitted waiting
Stopping / optional depth	Score the actual stopped readout for every declared target	Discovery, rejected proposals and executed continuations
Silence reception	Explain how queries read a pending or empty receiver	Timers, cancellation, retained state and readout
Online / structural update	Predict before that outcome; preserve feedback and version rules	Credit, replay, migration and optimizer
Random execution	Declare repeats; pair quality and cost from the same realization	Distributions/tails when budgets or deadlines matter

A simple warning: answering only the easy half can report zero conditional loss at 50% coverage, while full-workload loss with a unit fallback penalty remains 0.5. Selective prediction needs its own coverage/risk protocol. This is a semantic example, not a discovered error in saved language scores.

For finite choices, route credit is pi_i times (Q_i minus mean Q), where Q_i is the full conditional task/resource return. Immediate value utility omits unmodeled future writes, deadlines and work. More alternatives do not correct a wrong utility definition. Modeled time is not measured hardware latency.

A shared substrate opens input, communication and embodied opportunities

The opportunity register describes six application/mechanism directions with precedents, status and first proof conditions. The diagram illustrates a proposed event-native codec: learned memory and send/wait policies serve fixed reconstruction or task queries.

Irregular inputs can preserve time and recruit local programs. Quantized asynchronous outputs could allocate communication where useful. Count actual content, address, timing, framing and reset bits, plus both endpoints' work; variable message count is not itself a bitrate result.

Meta-learning offers fast-state/slow-rule mechanisms for retention, retrieval, writing and scheduling. Shared sensor/action/language paths could let embodied experience teach cognitive representations and reasoning guide action. Measure each transfer direction on held-out tasks with matched data/work and disconnected-path controls.

Existing variable-rate codecs, irregular-stream models and vision-language-action systems are relevant precedents. Our opportunity is their useful temporal/selective integration. No project codec, dexterity-transfer, AGI, customer or hardware-energy result follows from this chapter. Existing integrated queues retain priority.

Evidence supports the family; integration sets the agenda

Completed native results support content-bearing temporal computation, useful route credit and capacity beyond selected writes. The rows below share T256 and the same restricted 1M test region; each is single-seed. The 90M row changes data and width.

Native member	Fit M	BPC	Fit TFLOPs	MFLOPs/target
P32 / U2, timing-only credit	9.994	2.5064	7.221	0.722
P32 / U2, credit	9.994	2.3715	7.243	0.725
P32 / U4, credit	9.994	2.3452	11.950	1.196
P64 / U2, 90M data	89.997	1.8573	241.219	2.680

Saved arithmetic plus unit-counted special functions; eager full-step estimates. Whole-fit and per-fitting-target columns have consistent units across rows. Evaluation, setup, traffic and energy are additional.

Value choice credit improves the matched 10M D4/P32/U2 result by 0.134895 BPC for about 0.30% extra estimated fitting work. U2→U4 adds useful capacity with the same eight writes, but increases fitting work and untied parameters. These support scoped mechanisms; they are not modern frontier or public full-test wins.

Next integrated proof points: complete future-write utility and useful horizons; discovery/reception/silence; trained sparse-state and quality parity; matched repeated controls and measured traffic/energy. Native TTT, automatic structural design and shared-world integration require their own evidence.

The investment thesis is a reusable model/runtime/compute substrate from datacenters to mobile and robotics. The family definition makes that breadth concrete; commercial advantage remains an empirical quality/resource and adoption question.